



TSMC018 PDK usage guide:

An introduction on the usage of TSMC process design kits (PDK)

PDK Version: v1.1c Feb. 08th, 2010

TSMC018 PDK Usage Guide

● Introduction:

- This document describes the TSMC process design kits (PDK) parameterized cell (Pcell) software, which provides a graphical user interface that lets user create parameterized cells for placement in design layout.
- It is assumed that the user is familiar with the development and design of integrated circuits and with the cadence Virtuoso Layout Editor.
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● Overview:

■ The Symbol Display Information.

This section describes the symbol display information include four terminal NMOS symbol, four terminal PMOS symbol, three terminal npn BJT symbol, three terminal pnp BJT symbol, two terminal diode symbol, two terminal resistor symbol, three terminal resistor symbol, three terminal varactor symbol and two terminal varactor symbol.

■ Device Table

This section show the total device in this PDK. The user can check the page number in the device table to find out the CDF parameter and Pcell function.

- MOS Parameterized Cell Function Introduction
- BJT Parameterized Cell Function Introduction
- Diode Parameterized Cell Function Introduction
- Resistance Parameterized Cell Function Introduction
- Inductor Parameterized Cell Function Introduction
- Varactor Parameterized Cell Function Introduction
- Capacitor Parameterized Cell Function Introduction
- CDF Parameter Description

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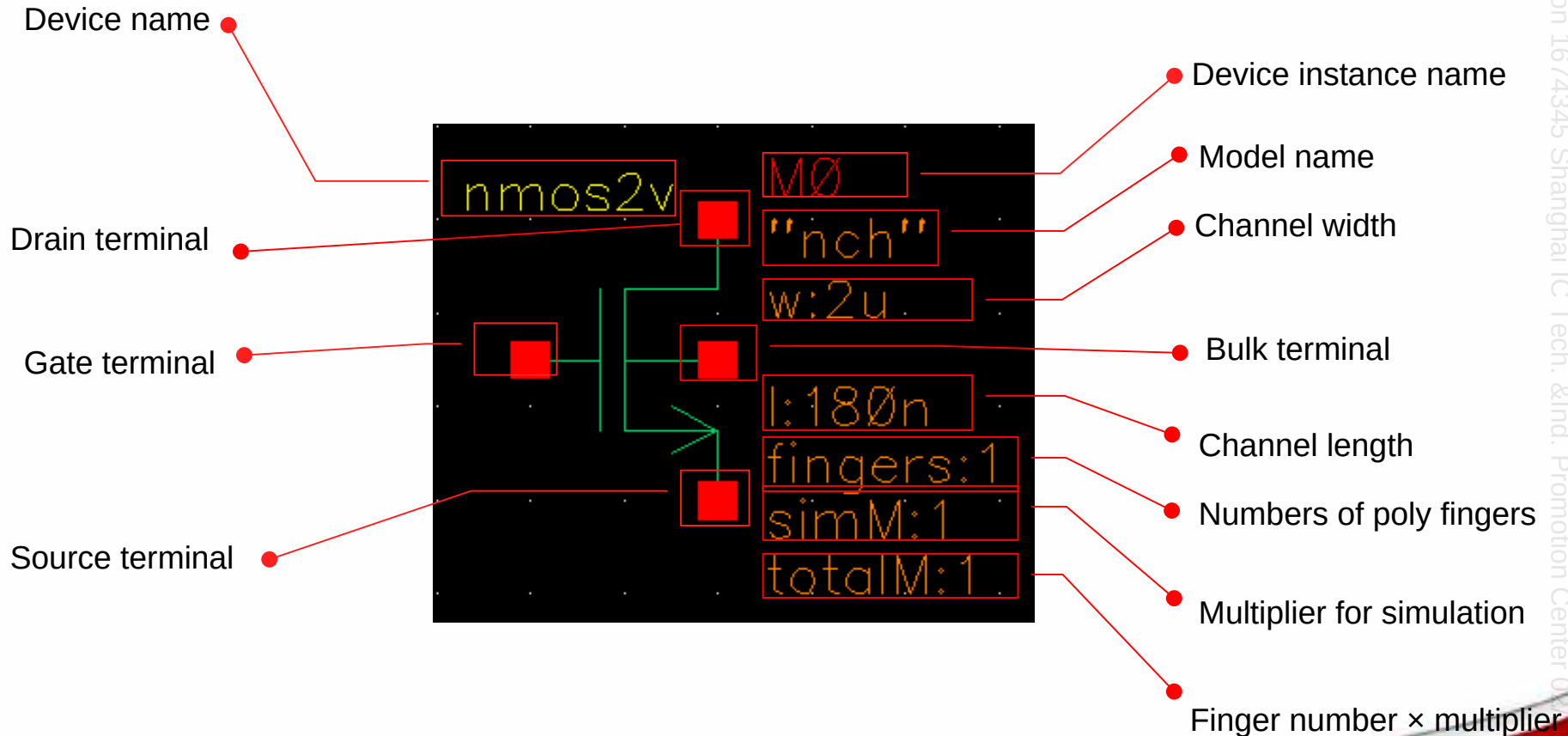
● Appendix:

- Appendix A – Abutment [Fast link](#)
- Appendix B – Stretch Handles [Fast link](#)

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● The Symbol Display Information:

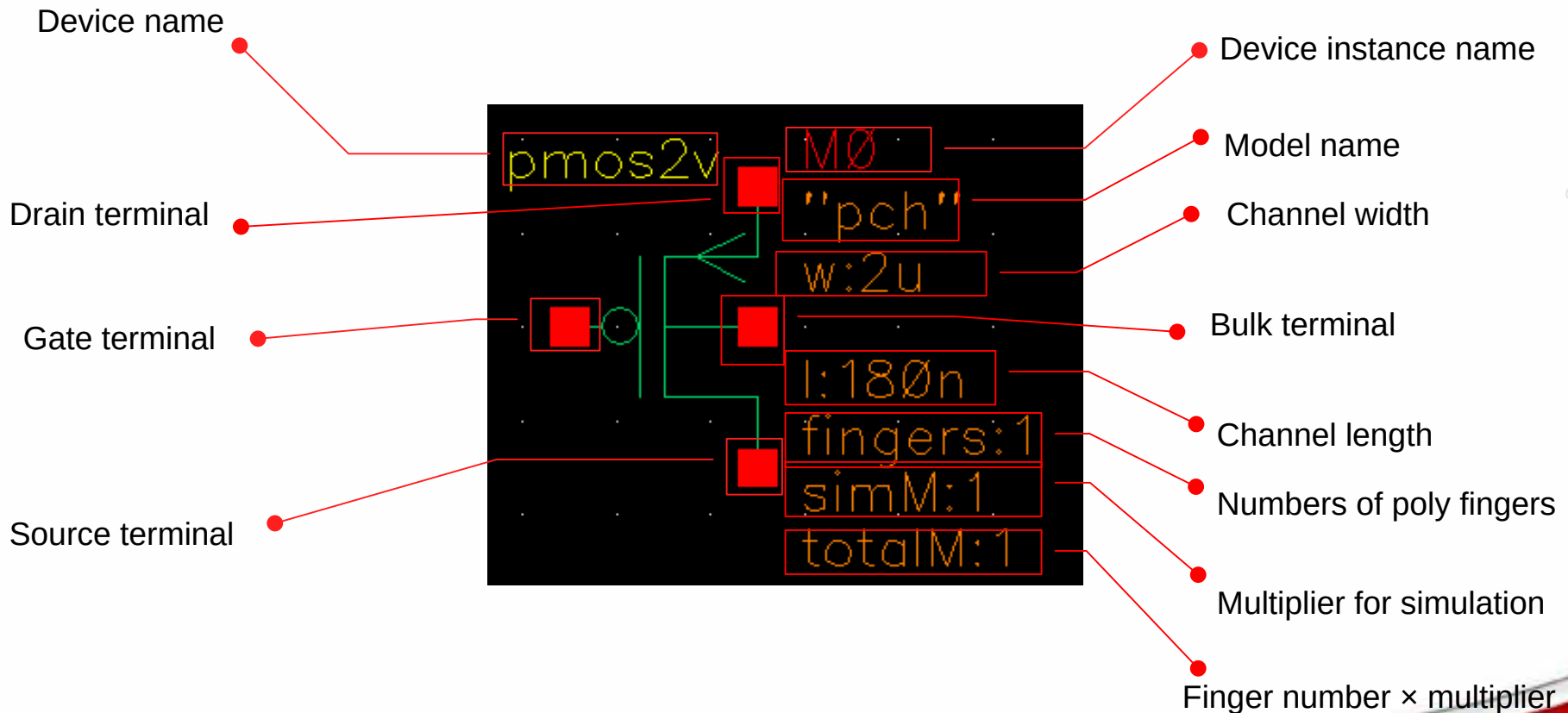
- The following figure shows the symbol for a four terminal NMOS:



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- The Symbol Display Information:

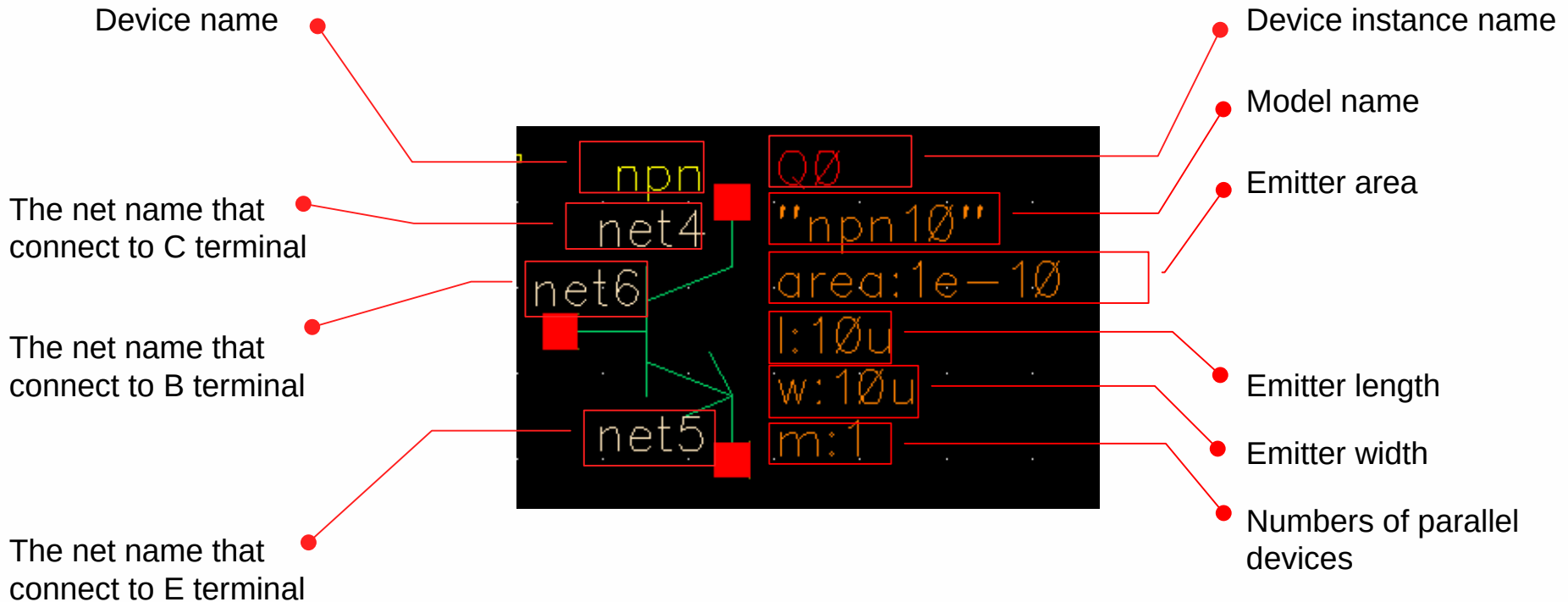
- The following figure shows the symbol for a four terminal PMOS:



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● The Symbol Display Information:

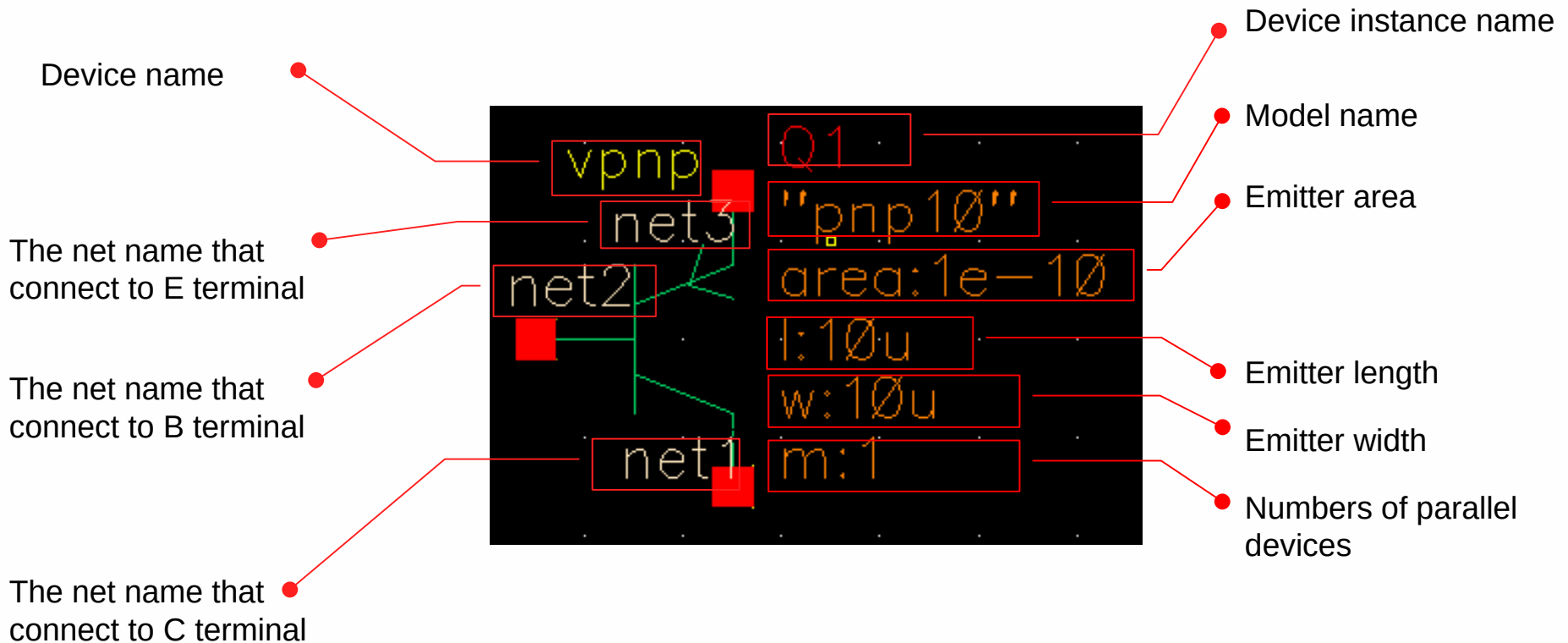
- The following figure shows the symbol for a three terminal npn BJT:



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● The Symbol Display Information:

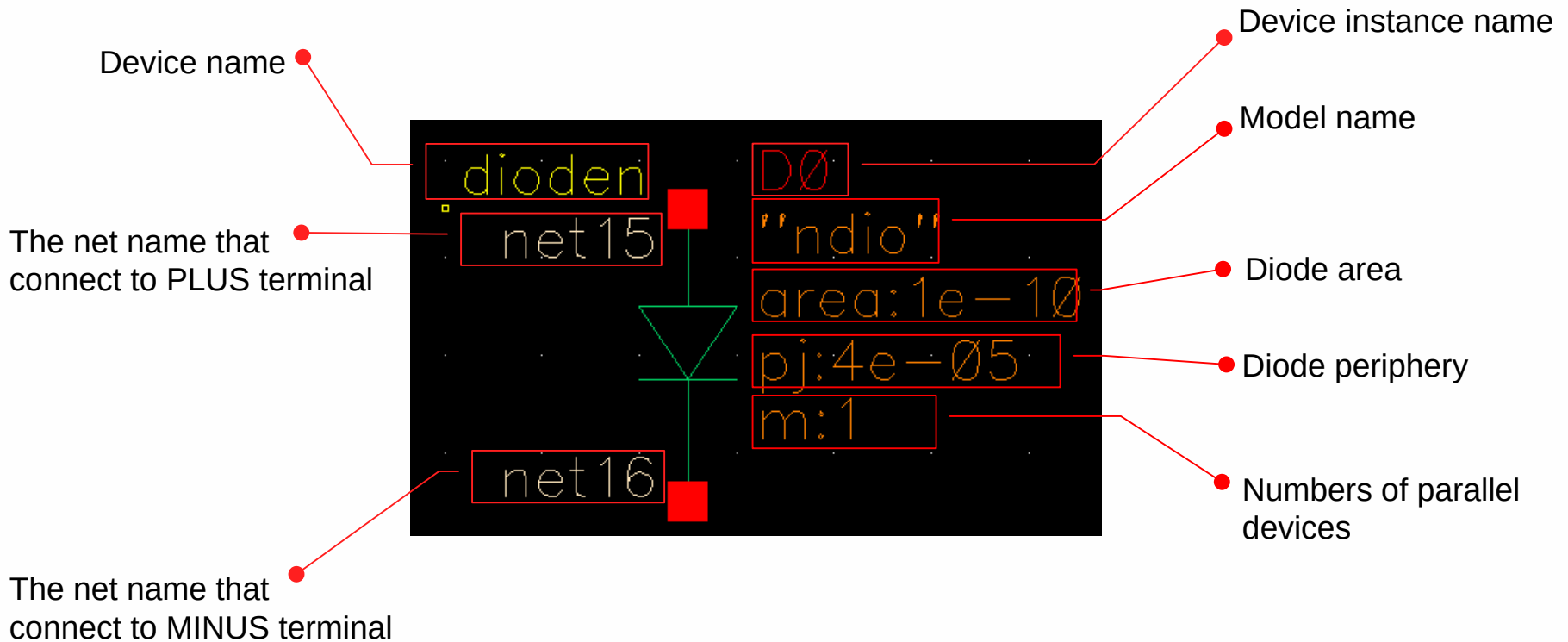
- The following figure shows the symbol for a three terminal pnp BJT:



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● The Symbol Display Information:

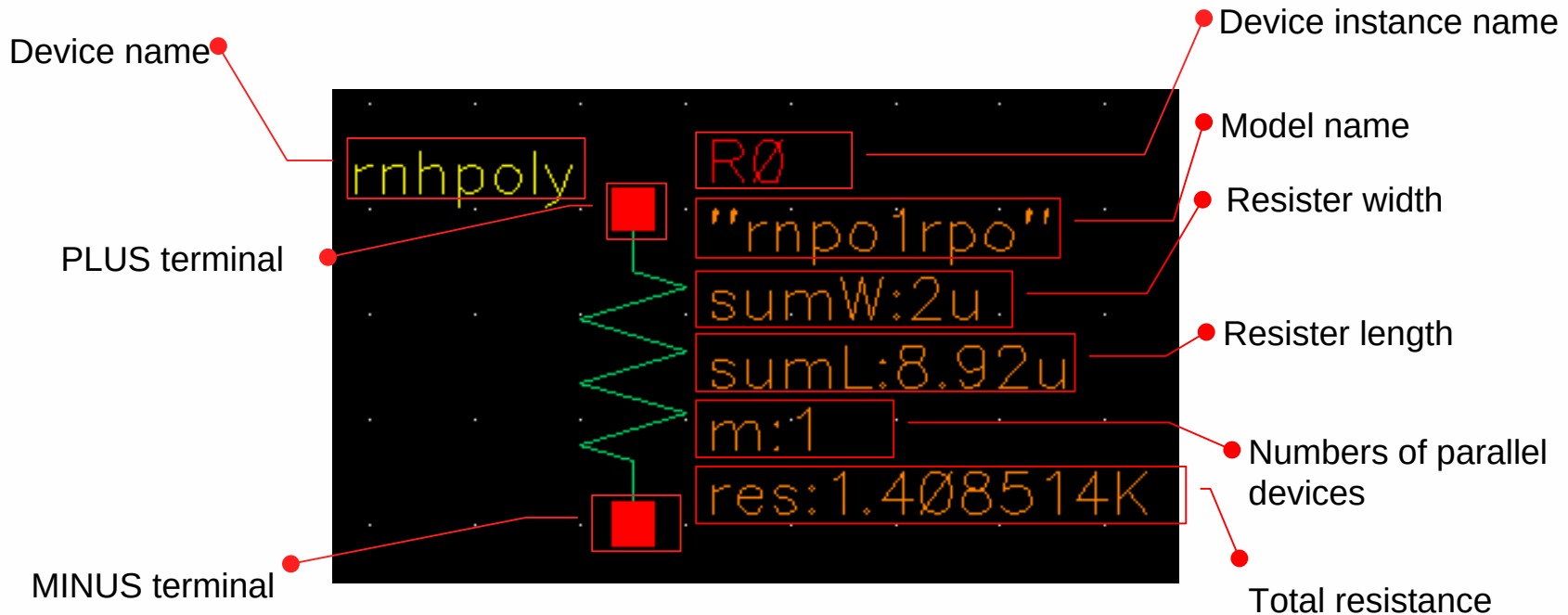
- The following figure shows the symbol for a two terminal diode:



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● The Symbol Display Information:

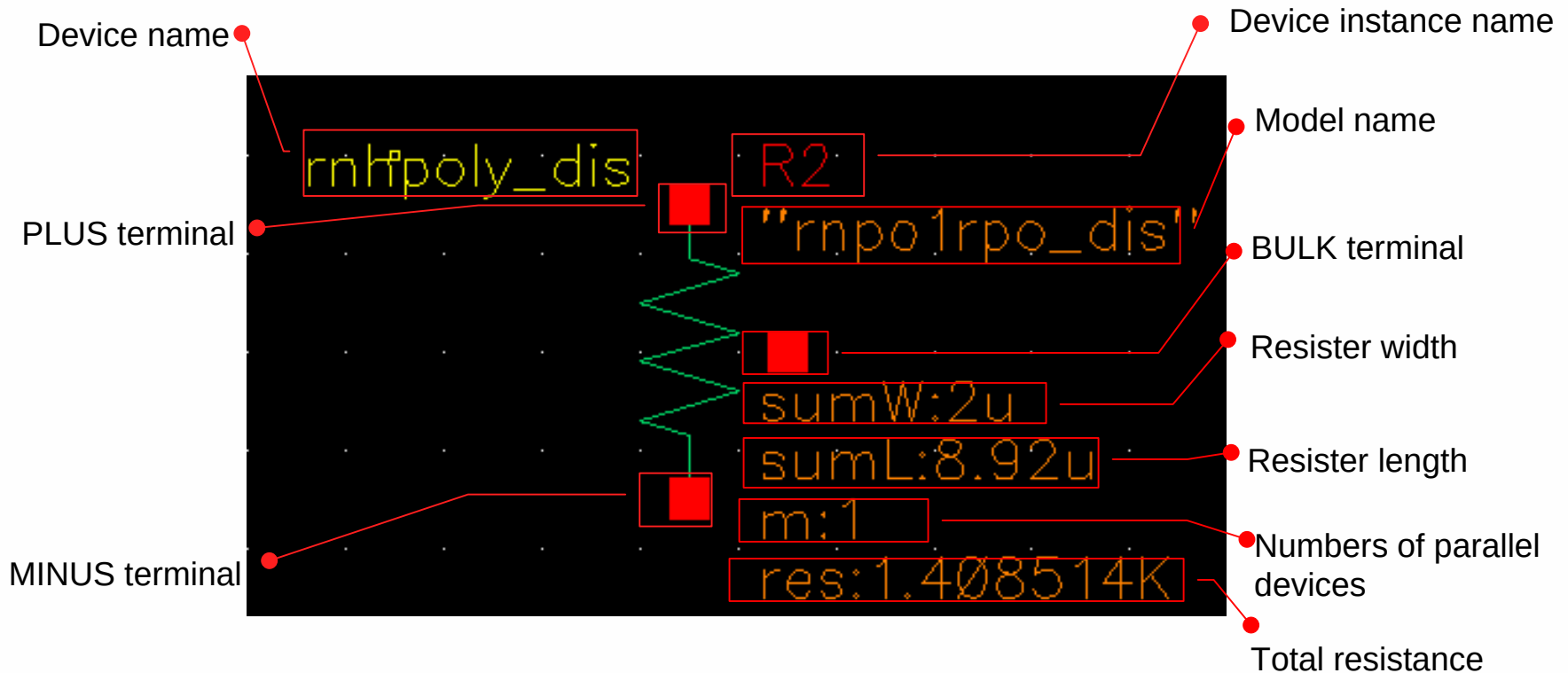
- The following figure shows the symbol for a two terminal resister:



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● The Symbol Display Information:

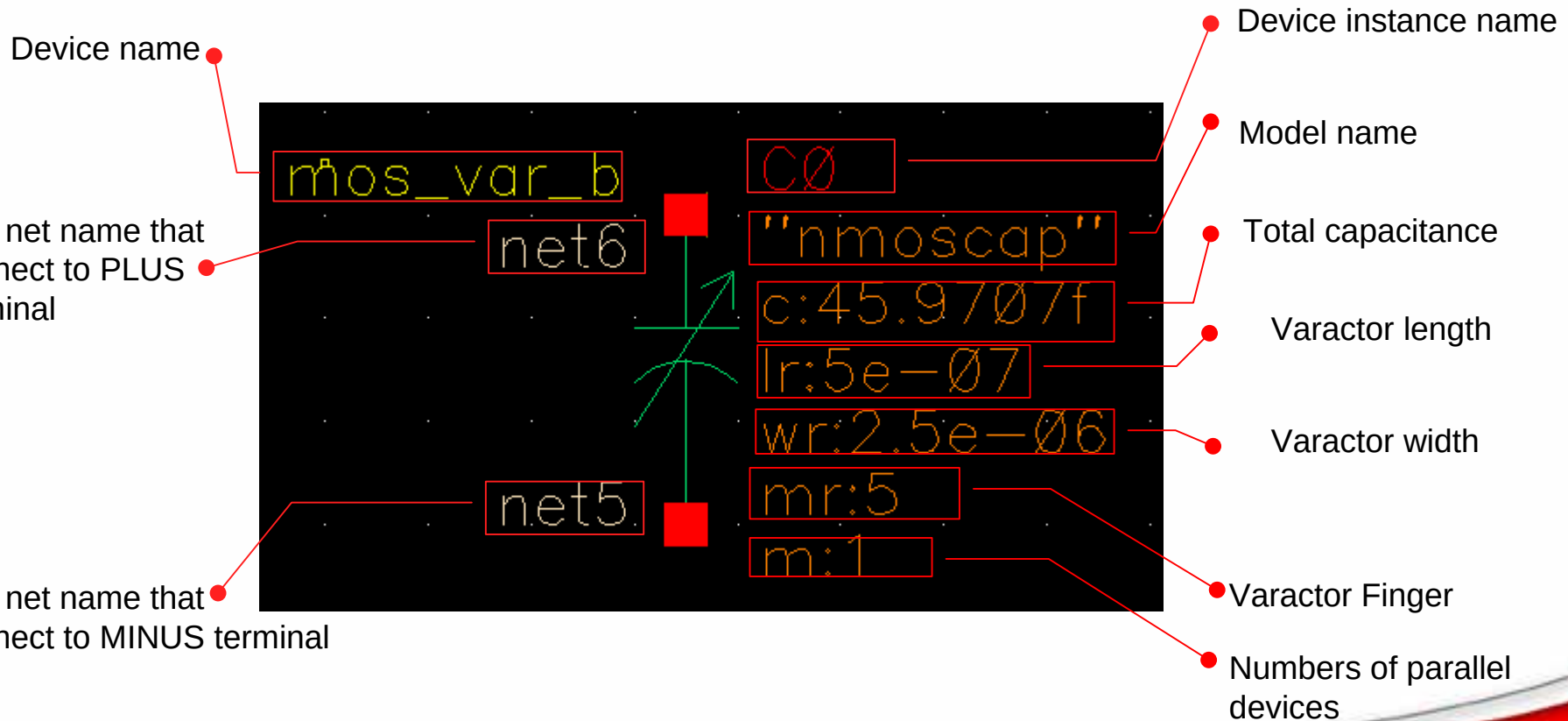
- The following figure shows the symbol for a three terminal resister:



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● The Symbol Display Information:

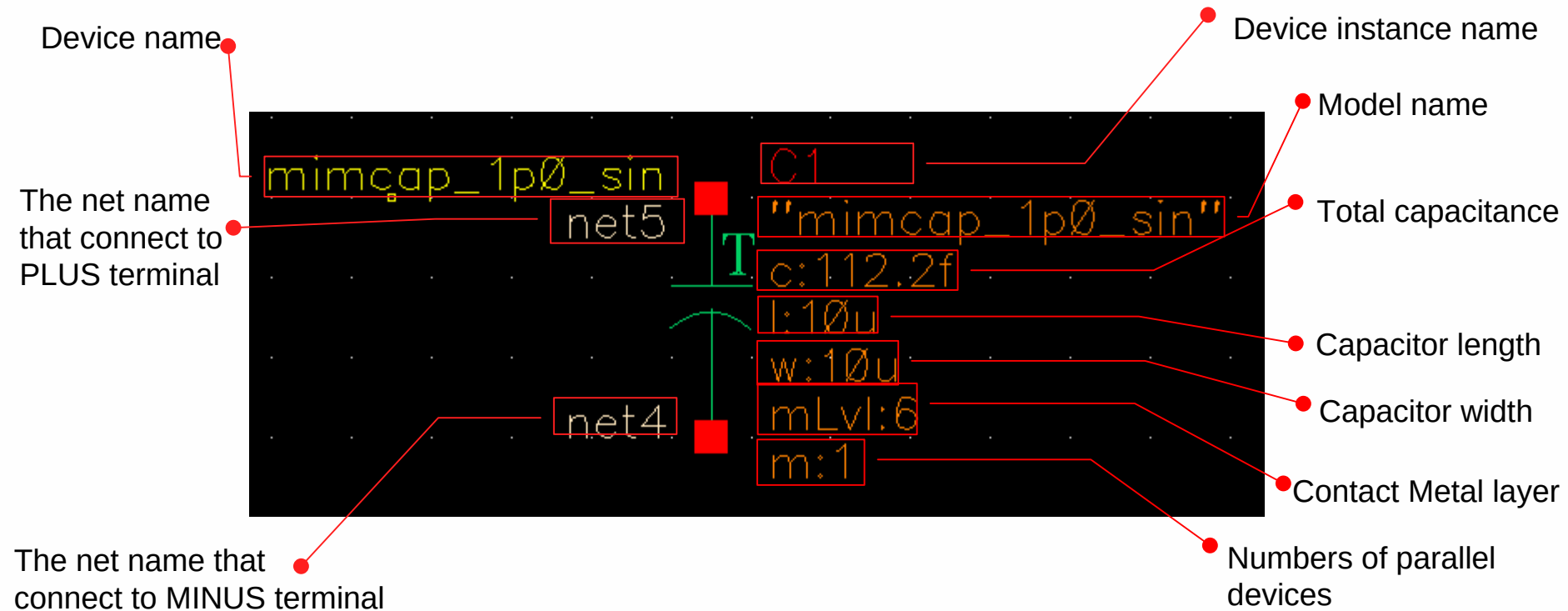
- The following figure shows the symbol for a two terminal varactor:



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● The Symbol Display Information:

- The following figure shows the symbol for a two terminal capacitor:



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● Device Table:

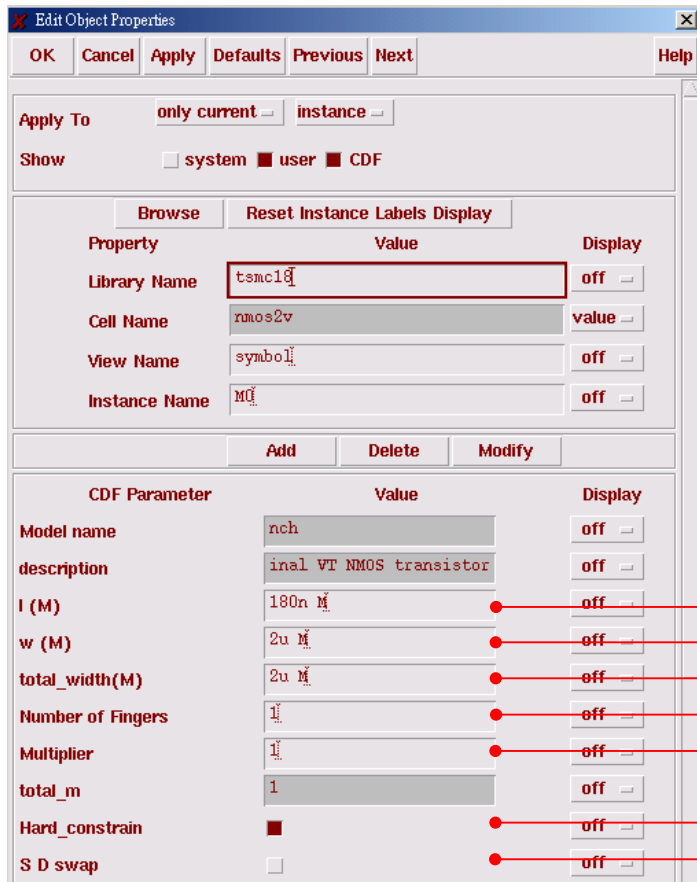
- The devices in this PDK are list as below table:

Categories	Device
MOS	nmos2v,nmos2v_mac,nmos2vdnw,nmos3v,nmos3v_mac,nmos3vdnw,nmosmvt2v,nmosmvt2v_mac,nmosmvt3v,
	nmosmvt3v_mac,nmosnvt2v,nmosnvt2v_mac,nmosnvt3v,nmosnvt3v_mac,pmos2v,pmos2v_mac,pmos3v,
	pmos3v_mac,pmosmvt2v,pmosmvt2v_mac,nmos2vdnw_mac,nmos3vdnw_mac
BJT	nnp,nnp_mis,vnpnp,vnpnp3,vnpnp_mis
Diode	dioden,dioden3v,diodenw,diodenw3v,diodep,diodep3v,ndio_m,ndio_3m,pdio_m
Resistor(1)	rm1,rm2,rm3,rm4,rm5,rmt
Resistor(2)	rnhpoly,rnhpoly_dis,rnlplus,rnlpoly,rnlpoly_dis,rnplplus,rnwell,rnwod,rnwsti_m,
	rphpoly,rphpoly_dis,rphripoly,rphripoly_d,s,rplplus,rplpoly,rplpoly_dis,
	rpplplus,rpplplus_2t,rnplplus_2t,rnlplus_2t,rnwod_2t,rpplplus_2t
Capacitors	crtmom,mimcap_1p0_sin,mimcap_1p0_sin_3t
Varactors	mos_var_b,mos_var_b3

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● MOS Parameterized Cell Function Introduction:

- The **schematic** component description format (CDF) parameter in MOS are list as below:



Property	Value	Display
Library Name	tsmc18	off
Cell Name	nmos2v	value
View Name	symbol	off
Instance Name	M0	off

CDF Parameter	Value	Display
Model name	nch	off
description	inal VT NMOS transistor	off
L (M)	180n	off
W (M)	2u	off
total_width(M)	2u	off
Number of Fingers	1	off
Multiplier	1	off
total_m	1	off
Hard_constrain	☒	off
S D swap	☐	off

◆ **Model name:** Display Model name information.

◆ **Description:** Display device description.

(These parameters can't be modify in CDF form)

◆ **L (M):** Channel length of the device.

◆ **W (M):** Channel width of the device.

◆ **Total_width(M):** Total channel width of this device, equal to width x fingers.

◆ **Number of Fingers_(N):** Numbers of poly fingers.

[Check here for more information](#)

◆ **Multiplier:** Numbers of parallel MOS device.

[Check here for more information](#)

◆ **Total_m:** Display numbers of parallel MOS device.

(This parameter can't be modify in CDF form)

◆ **Hard_constrain:** This function provides an option to constrain the value for each parameter in this device.

◆ **S D swap:** Enable this function to swap source and drain terminal.

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● MOS Parameterized Cell Function Introduction:

- The **schematic** component description format (CDF) parameter in MOS are list as below:

Front page

Calc Diff Params	☒	off
Calc SA SB SD	☒	off
Source_area	9.6e-13	off
Drain_area	9.6e-13	off
Source_periphery_(M)	4.96u M	off
Drain_periphery_(M)	4.96u M	off
NRS	0.135	off
NRD	0.135	off
SA(LOD_effect)_(M)	480.0n M	off
SB(LOD_effect)_(M)	480.0n M	off
Well_Proximity_Effect	off	off
SCA (M)	0 M	off
SCB (M)	0 M	off
SCC (M)	0 M	off

◆ **Calc Diff Params:** The switch provide to modify simulation parameters.

◆ **Calc SA SB SD:** The switch provide to modify simulation SA, SB and SD.

◆ **Source_area:** Source area (AS) - for simulate use.

◆ **Drain_area:** Drain area (AD) - for simulate use.

◆ **Source_periphery:** Source periphery (PS) - for simulate use.

◆ **Drain_periphery:** Drain periphery (PD) - for simulate use.

◆ **NRS:** Number of squares source resistance – for simulate use.

◆ **NRD:** Number of squares drain resistance – for simulate use.

[Check here for more information](#)

◆ **SA(LOD_effect)_(M):** LOD effect parameter – for simulate use.

◆ **SB(LOD_effect)_(M):** LOD effect parameter – for simulate use.

◆ **Well Proximity Effect Cal. Method:**

two entry methods for WPE parameters:

off => use model default values.

custom => user input desire value by self.

◆ **SCA (M):** WPE SCA parameter. - for simulate use

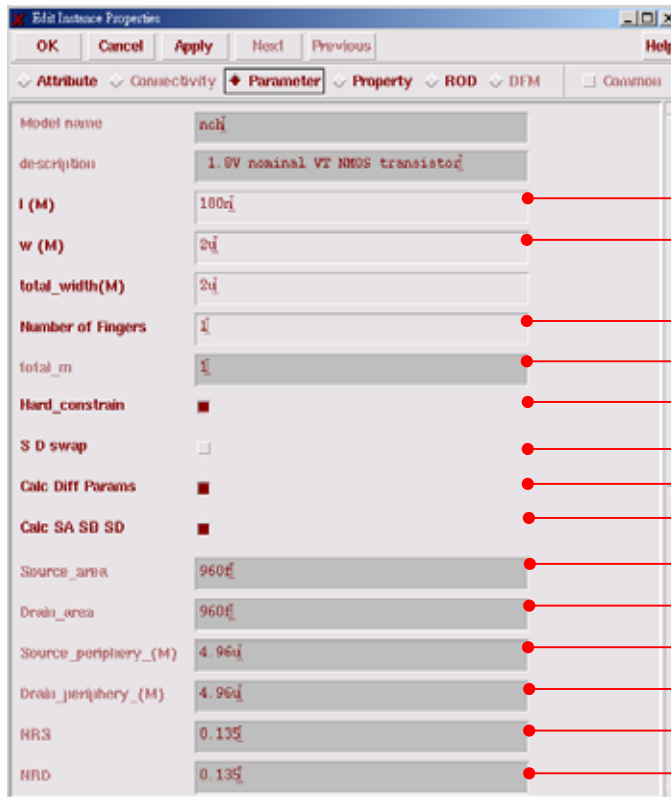
◆ **SCB (M):** WPE SCB parameter. - for simulate use

◆ **SCC (M):** WPE SCC parameter. - for simulate use

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● MOS Parameterized Cell Function Introduction:

- The **layout** component description format (CDF) parameter in MOS are list as below:



Parameter	Value
Model name	nom1
description	1.8V nominal VT NMOS transistor
L (M)	180nm
W (M)	2um
total_width(M)	2um
Number of Fingers	1
total_m	1
Hard_constrain	☒
S/D swap	☐
Calc Diff Params	☒
Calc SA SB SD	☒
Source_area	960um^2
Drain_area	960um^2
Source_periphery_(M)	4.96um
Drain_periphery_(M)	4.96um
NRS	0.135
NRD	0.135

- ◆ **Model name:** Display Model name information.
- ◆ **Description:** Display device description.
- ◆ **L (M):** Channel length of the device.
- ◆ **W (M):** Channel width of the device.
- ◆ **Total_width(M):** Total channel width of this device, equal to width x fingers.
- ◆ **Number of Fingers_(N):** Numbers of poly fingers.
- ◆ **Total_m:** Display numbers of parallel MOS device.
- ◆ **Hard_constrain:** This function provides an option to constrain the value for each parameter in this device.
- ◆ **S D swap:** Enable this function to swap source and drain terminal.
(These are the same parameter that in schematic CDF form)
- ◆ **Calc Diff Params:** The switch provide to modify simulation parameters.
- ◆ **Calc SA SB SD:** The switch provide to modify simulation SA, SB and SD.
- ◆ **Source_area:** Source area (AS) - for simulate use.
- ◆ **Drain_area:** Drain area (AD) - for simulate use.
- ◆ **Source_periphery:** Source periphery (PS) - for simulate use.
- ◆ **Drain_periphery:** Drain periphery (PD) - for simulate use.
- ◆ **NRS:** Number of squares source resistance – for simulate use.
- ◆ **NRD:** Number of squares drain resistance – for simulate use.
(These are the same parameter that in schematic CDF form)

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● MOS Parameterized Cell Function Introduction:

- The **layout** component description format (CDF) parameter in MOS are list as below:

Front page

SA(LOD_effect)_(M)	480.0n
SB(LOD_effect)_(M)	480.0n
SD(Fingers_Spacing)_(M)	540.0n
fingers_SP_INC(M)	0
Well_Proximity_Effect	off
SCA (M)	0
SCB (M)	0
SCC (M)	0
leftCnt	■
rightCnt	■
interCnt	■
routePolydir	None
route_Source_Drain	None
bodytie_typeL	None
bodytie_typeR	None
Text size	0.05
Imp layer	■

◆SA(LOD_effect)_(M): LOD effect parameter – for simulate use.

◆SB(LOD_effect)_(M): LOD effect parameter – for simulate use.

◆fingers_SP_INC(M): The function is provided to modify poly gate space.

The function only appears when Number of fingers_(N) > 1

[Check here for more information](#)

◆leftCnt: A option for drawing poly-left diffusion area metal1 connection.

◆rightCnt: A option for drawing poly-right diffusion area metal1 connection.

[Check here for more information](#)

◆routePolydir:(None, Top, Bottom, Both) A option for drawing poly gate connection.

The function only appears when Number of fingers_(N) > 1

[Check here for more information](#)

◆route_Source_Drain:(Source, Drain, Both) A option for drawing source and drain connection.

The function only appears when Number of fingers_(N) > 1

[Check here for more information](#)

◆bodytie_typeL: (None, Integrated, Detached) A option for drawing body connection.

◆bodytie_typeR: (None, Integrated, Detached) A option for drawing body connection.

[Check here for more information](#)

◆Text size: The function can modify the font value in layout view.

◆Imp layer: The function provide a option for well implant.

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● BJT Parameterized Cell Function Introduction:

- The **schematic** component description format (CDF) parameter in BJT are list as below:

Property	Value	Display
Library Name	tsmc18	off
Cell Name	nnp	value
View Name	symbol	off
Instance Name	Q1	off

CDF Parameter	Value	Display
Model name	nnp10	off
description	1.8V NPN BIPOLAR	off
EmitterSize	10x10	off
EmitterArea	1e-10	off
Multiplier	1	off
Hard_constrain	☒	off
IxUseCell	nnp10	off

◆ **Model name:** Display model name information.

◆ **Description:** Display device description.

(These parameter can't be modify in CDF form)

◆ **EmitterSize:** (2X2, 5X5, 10X10) Select the bjt dimension in design layout.

[Check here for more information](#)

◆ **EmitterArea:** Display the bjt emitter area.

(This parameter can't be modify in CDF form)

◆ **Multiplier:** Numbers of parallel MOS device.

[Check here for more information](#)

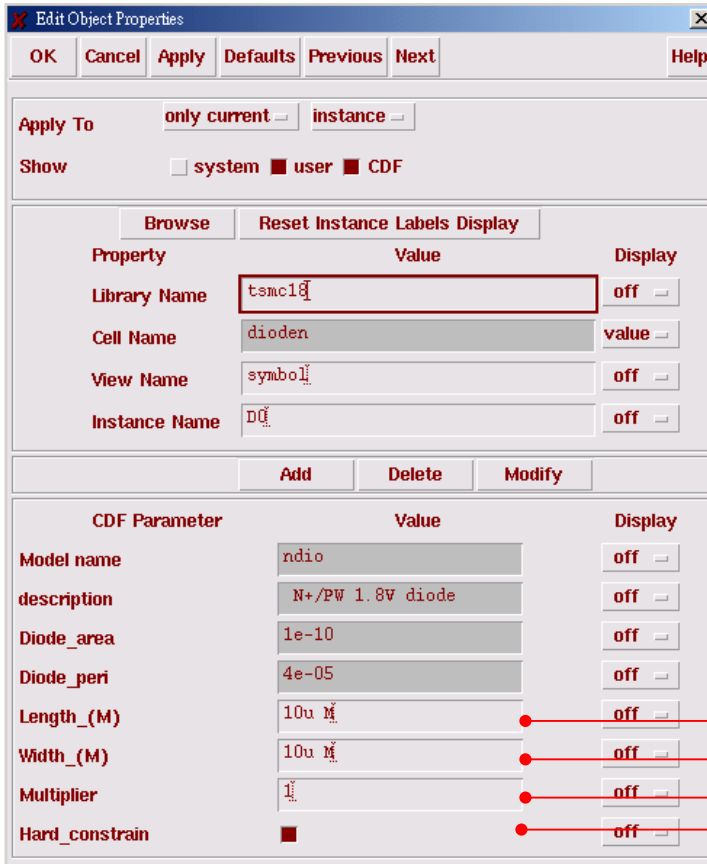
◆ **Hard_constrain:** This function provides an option to constrain the value for each parameter in this device.

◆ **IxUseCell:** Display the layout cell name.

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● Diode Parameterized Cell Function Introduction:

- The **schematic** component description format (CDF) parameter in Diode are list as below:



Property	Value	Display
Library Name	tsmc18	off
Cell Name	diode	value
View Name	symbol	off
Instance Name	D1	off

CDF Parameter	Value	Display
Model name	ndio	off
description	N+/PW 1.8V diode	off
Diode_area	1e-10	off
Diode_peri	4e-05	off
Length_(M)	10u M	off
Width_(M)	10u M	off
Multiplier	1	off
Hard_constrain	☒	off

◆ **Model name:** Display model name information.

◆ **Description:** Display device description.

◆ **Diode_area:** Display the diode area.

◆ **Diode peri:** Display the diode periphery.

(These parameter can't be modify in CDF form)

◆ **Length_(M):** Junction length of the device.

◆ **Width_(M):** Junction Width of the device.

◆ **Multiplier:** Numbers of parallel Diode device.

[Check here for more information](#)

◆ **Hard_constrain:** This function provides an option to constrain the value for each parameter in this device.

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● Resistance (1) Parameterized Cell Function Introduction:

- The **schematic** component description format (CDF) parameter in resistance are list as below:

Property	Value	Display
Library Name	tsmc18	off
Cell Name	rm1	value
View Name	symbol	off
Instance Name	R0	off

CDF Parameter	Value	Display
Model name	rm1	off
Total resistance(ohms)	89.7m Ohms	off
LVS Resistance(ohms)	78m Ohms	off
Width(M)	230n M	off
Length(M)	230n M	off
Multiplier	1	off
Rsh(ohms/square)	78m	off
Res_update_method	I & W	off
Hard_constrain	☒	off

◆ **Model name:** Display model name information.

◆ **Total resistance(ohms) :** Display device description. .

◆ **LVS Resistance(ohms):** Device resistance value.
(These parameter can't be modify in CDF form)

◆ **Width(M):** Device segment width.

◆ **Length(M):** Device segment length.

◆ **Multiplier:** Numbers of parallel Diode device.

[Check here for more information](#)

◆ **Rsh(ohms/square):** Display the device Rs value.
(This parameter can't be modify in CDF form)

◆ **Res_update_method:** (I & W, Res & W) Res update method, please review **segment width** and **segment length** function.

◆ **Hard_constrain:** This function provides an option to constrain the value for each parameter in this device.

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● Resistance(2) Parameterized Cell Function Introduction:

- The **schematic** component description format (CDF) parameter in resistance are list as below:

Property	Value	Display
Library Name	tsmc18	off
Cell Name	rnhpoly	value
View Name	symbol	off
Instance Name	R1	off

CDF Parameter	Value	Display
Model name	rnpolrpo	off
description	sistor without silicide	off
Total resistance(ohms)	1.408514K Ohms	off
Total width(M)	2u M	off
Segment width(M)	2u M	off
Total length(M)	8.92u M	off
Segment length(M)	8.92u M	off
Multiplier	1	off
Rs(ohms/square)	295.3	off
Resistor connection	Series Parallel	off
Number of segments	1	off
Segment spacing(M)	250n M	off
Cont columns	1	off
With Mismatch Effect		off
Hard constrain		off
Res_update_method	I & W	off

◆ **Model name:** Display model name information.

◆ **Description:** Display device description.

(These parameter can't be modify in CDF form)

◆ **Total resistance(ohms):** Device resistance value.

◆ **Segment width(M):** Device segment width.

◆ **Segment length(M):** Device segment length.

[Check here for more information](#)

◆ **Total width(M):** Display the device segment width.

◆ **Total length(M):** Display the device segment length.

◆ **Rs(ohms/square):** Display the device Rs value.

(These parameter can't be modify in CDF form)

◆ **Multiplier:** Numbers of parallel Diode device.

[Check here for more information](#)

◆ **Resistor connection:** Device resistance value.

◆ **Number of segment:** Device segment width.

◆ **Segment spacing(M):** Device segment length.

[Check here for more information](#)

◆ **Cont columns:** Device contact columns number.

[Check here for more information](#)

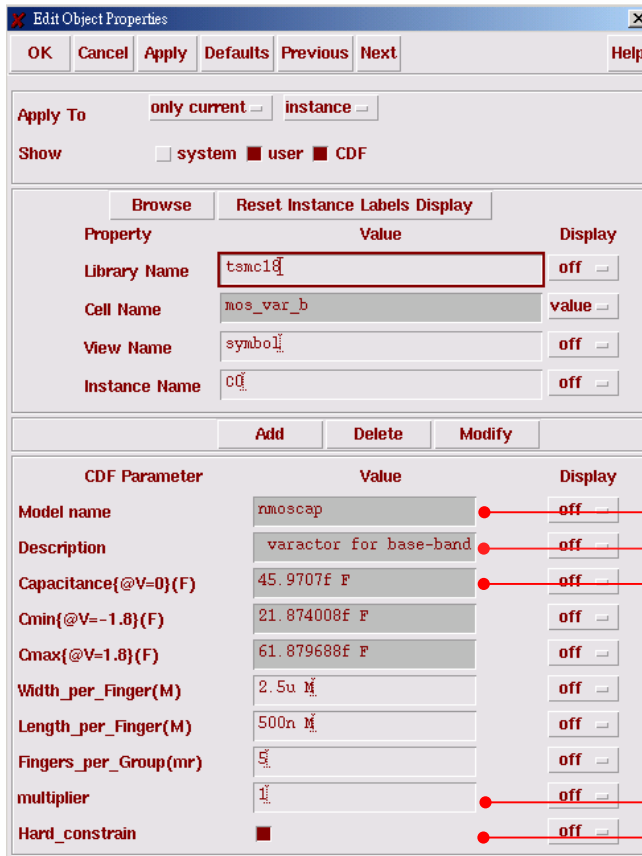
◆ **Hard constrain:** This function provides an option to constrain the value for each parameter in this device.

◆ **Res_update_method:** (I & W, Res & W) Res update method, please review [segment width](#) and [segment length](#) function.

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● Varactor Parameterized Cell Function Introduction:

- The **schematic** component description format (CDF) parameter in varactor are list as below:



Property	Value	Display
Library Name	tsmc18	off
Cell Name	mos_var_b	value
View Name	symbol	off
Instance Name	c0	off

CDF Parameter	Value	Display
Model name	nmoscip	off
Description	varactor for base-band	off
Capacitance{@V=0}(F)	45.9707f F	off
Cmin{@V=-1.8}(F)	21.874008f F	off
Cmax{@V=1.8}(F)	61.879688f F	off
Width_per_Finger(M)	2.5u M	off
Length_per_Finger(M)	500n M	off
Fingers_per_Group(mr)	5	off
multiplier	1	off
Hard_constrain	☒	off

- ◆ **Model name:** Display model name information.
- ◆ **Description:** Display device description.
- ◆ **Capacitance{@V=0}(F):** Display the capacitance value in 0V.
- ◆ **Cmin{@V=-vdd}(F):** Display the capacitance value in -vdd
- ◆ **Cmax{@V=vdd}(F):** Display the capacitance value in vdd
(These parameter can't be modify in CDF form)
- ◆ **Width_per_Finger(M):** Device width.
- ◆ **Length_per_Finger(M):** Device length.
- ◆ **Fingers_per_Group(mr):** Device finger numbers.
- ◆ **Multiplier:** Numbers of parallel Varactor device.
[Check here for more information](#)
- ◆ **Hard_constrain:** This function provides an option to constrain the value for each parameter in this device.

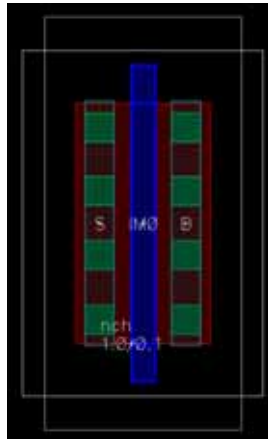
CDF Parameter Description

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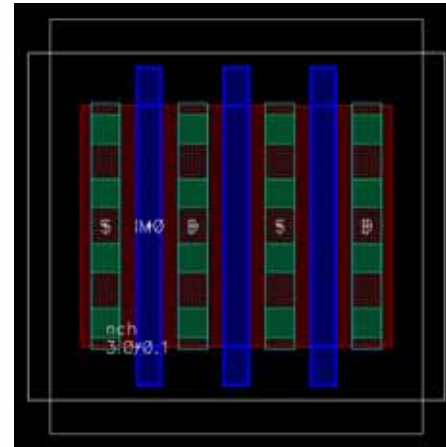
● The function of **Number of Fingers_(N)**

- This parameter provide user to increment the poly finger numbers.

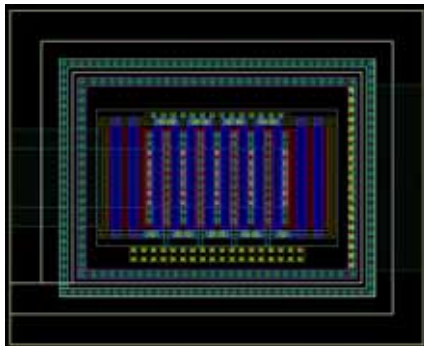
Number of Fingers_(N)=1



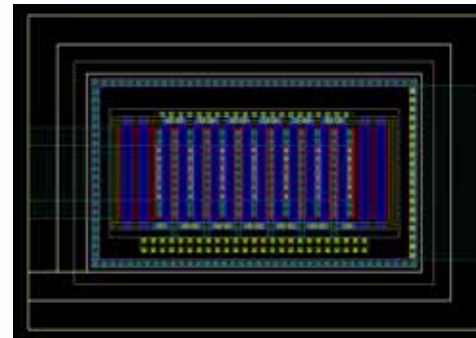
Number of Fingers_(N) =3



Number of Fingers_(N)=8



Number of Fingers_(N) =12



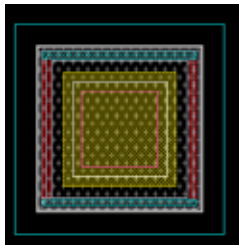
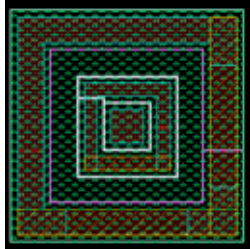
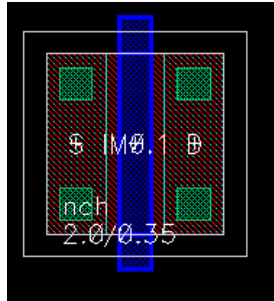
[Check here to back to MOS](#)

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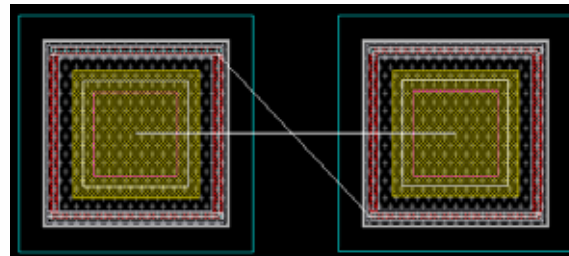
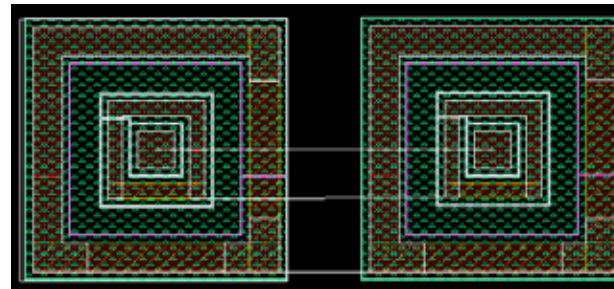
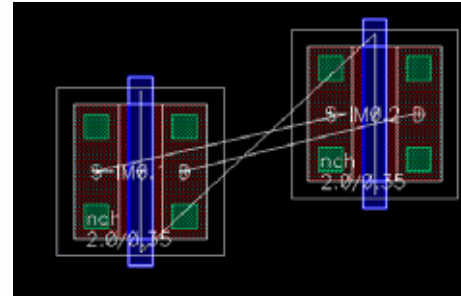
● The function of **Multiplier**

- This parameter provide user to increment the **parallel** device.

Multiplier = 1



Multiplier = 2



[Check here to back to MOS](#)

[Check here to back to BJT](#)

[Check here to back to Diode](#)

[Check here to back to Resistance\(1\)](#)

[Check here to back to Resistance\(2\)](#)

[Check here to back to Varactor](#)

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- The function of **Calc Diff Params** and **Calc SA SB SD**

- It's a switch for input simulation parameter that include area of source (AS), area of drain (AD), periphery of source (PS), periphery of drain (PD), number of squares source resistance (NRS), number of squares drain resistance (NRD) and LOD effect parameter- SA , SB and SD. Modify those parameters only influence simulation conditions, the design layout will not have any different.

Calc Diff Params is enable

Calc Diff Params	☒	off
Source_area	2.3e-13	off
Drain_area	2.3e-13	off
Source_periphery_(M)	2.46u M	off
Drain_periphery_(M)	2.46u M	off
NRS	0.13	off
NRD	0.13	off

Parameters **can't** be modify

Calc Diff Params is disable

Calc Diff Params	☐	off
Source_area	2.3e-13	off
Drain_area	2.3e-13	off
Source_periphery_(M)	2.46u M	off
Drain_periphery_(M)	2.46u M	off
NRS	0.13	off
NRD	0.13	off

Parameters **can** be modify

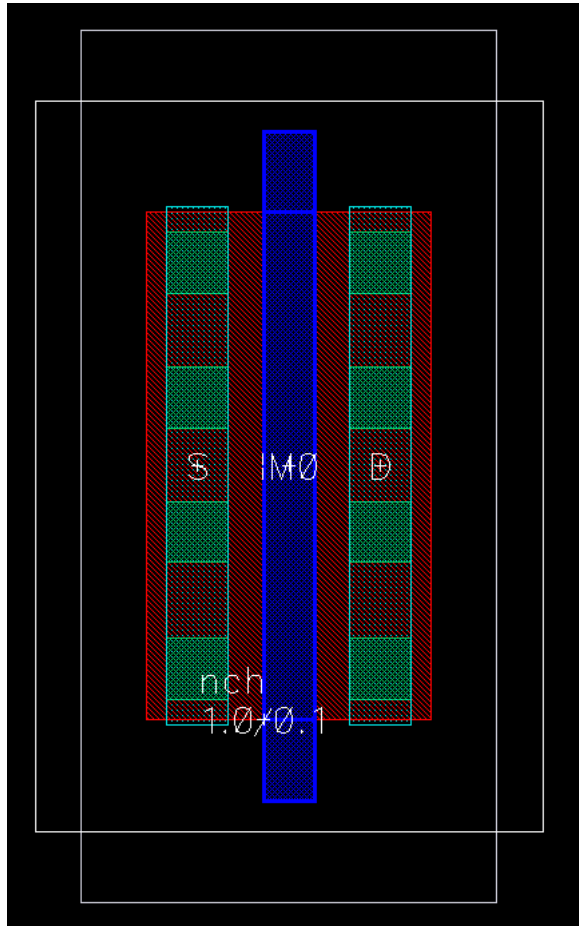
[Check here to back to MOS](#)

TSMC018 PDK Usage Guide

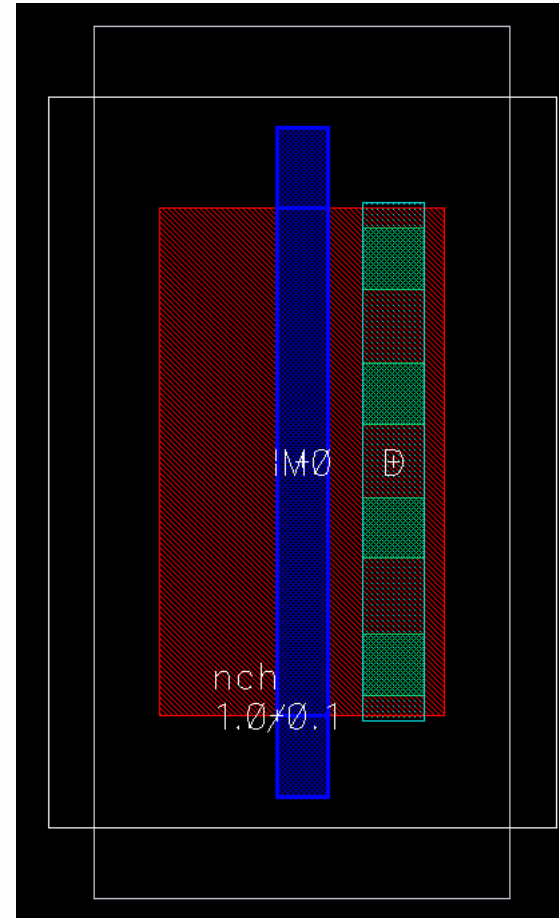
- The function of **leftCnt**, **RightCnt**

- The function provide a option for drawing poly-left (right) diffusion area metal1 connect

leftCnt is enable



leftCnt is Disable



[Check here to back to MOS](#)

TSMC018 PDK Usage Guide

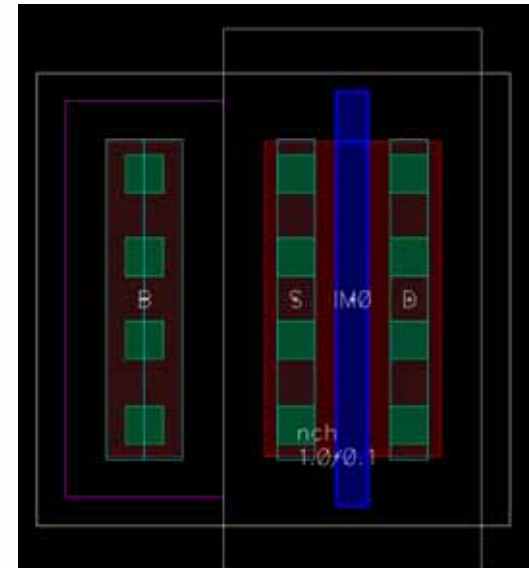
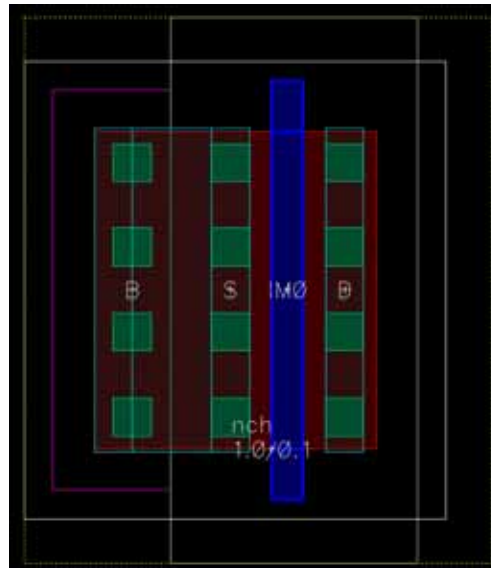
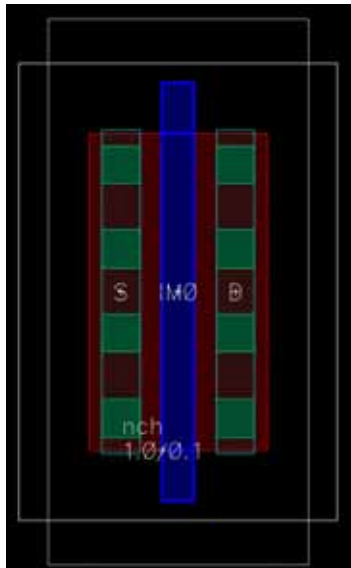
- The function of **bodytie_typeL** and **bodytie_typeR**

- The function provide a option for drawing body connection at the device left (**bodytie_typeL**) or device right (**bodytie_typeR**).

bodytie_typeL is *None*

bodytie_typeL is *Integred*

bodytie_typeL is *Detached*



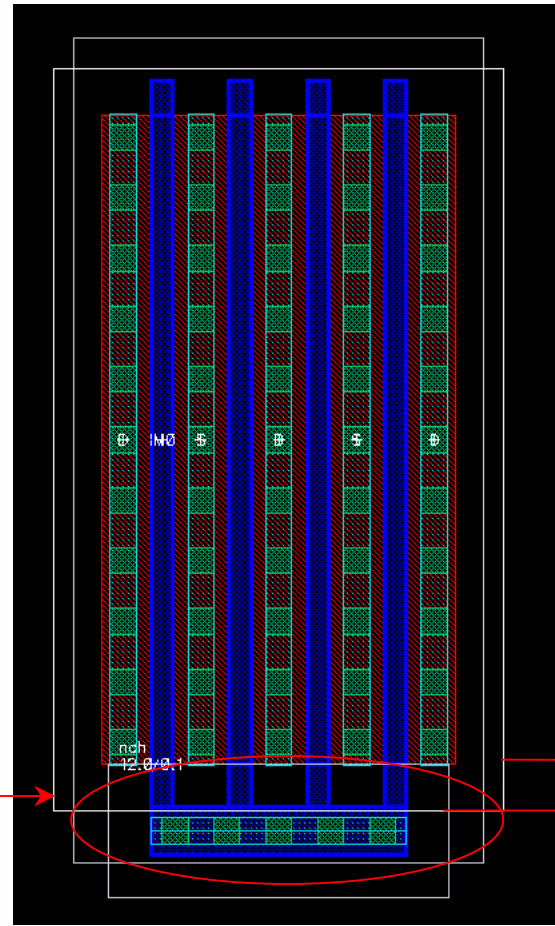
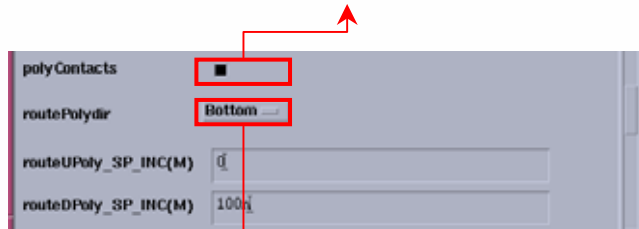
[Check here to back to MOS](#)

TSMC018 PDK Usage Guide

● The function of **routePolydir**

- The function is provided to drawing poly gate connection. The space of poly gate connection to the diffusion area can be modify by **routeUPoly_SP_INC(M)** and **routeDPoly_SP_INC(M)**².

The **poly Contacts** will appear when **routePolydir** doesn't None. It is an option to draw contact on the poly gate.



PO.S.6(design rule)
+ routeDPoly_SP_INC

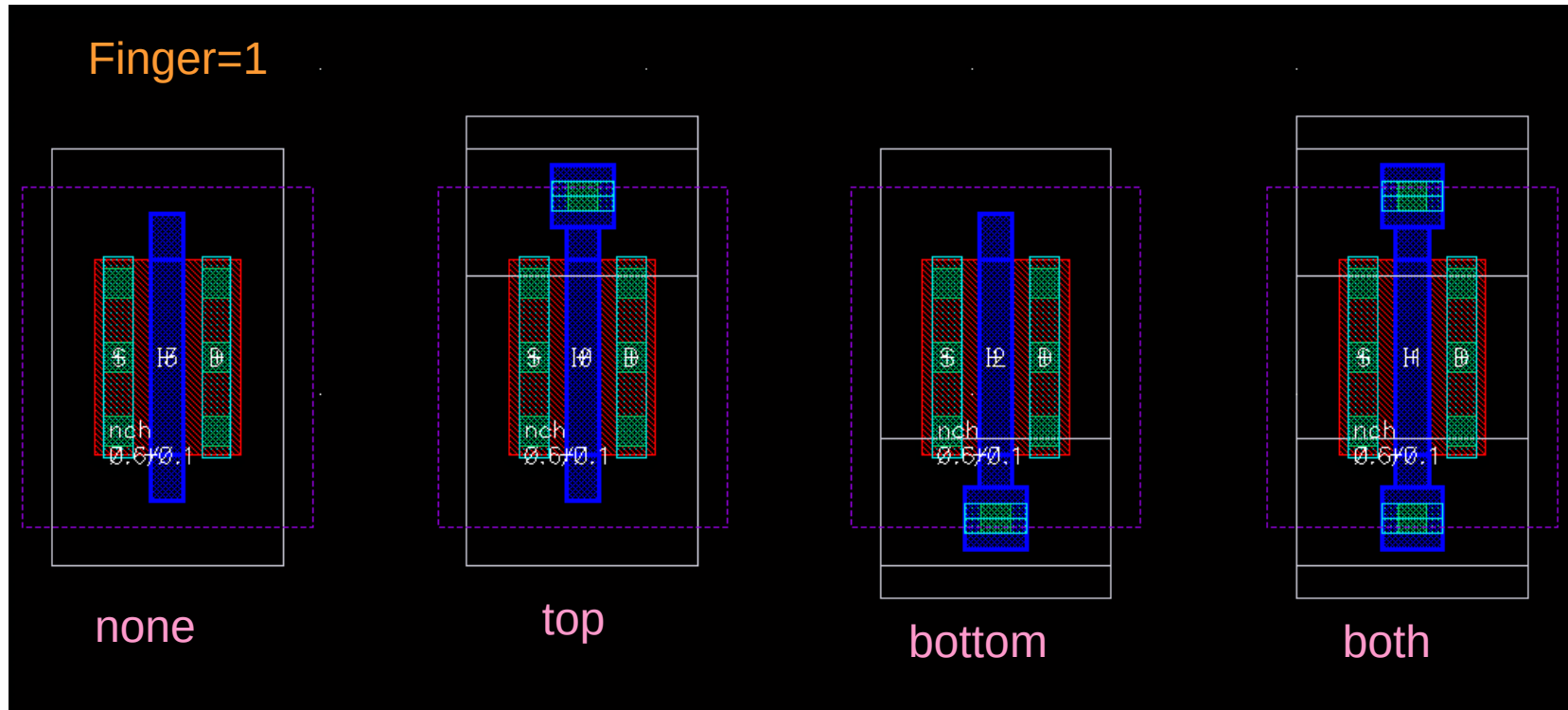
[Check here to back to MOS](#)

² The routeUPoly_SP_INC(M) and routeDPoly_SP_INC(M) only appear when routePolydir doesn't **None**.

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● The function of routePolydir

- The function is suggest to use in multi-finger. If customer use only one finger and turn on the routePolydir none/top/bottom/both direction. It will shows the following layout.

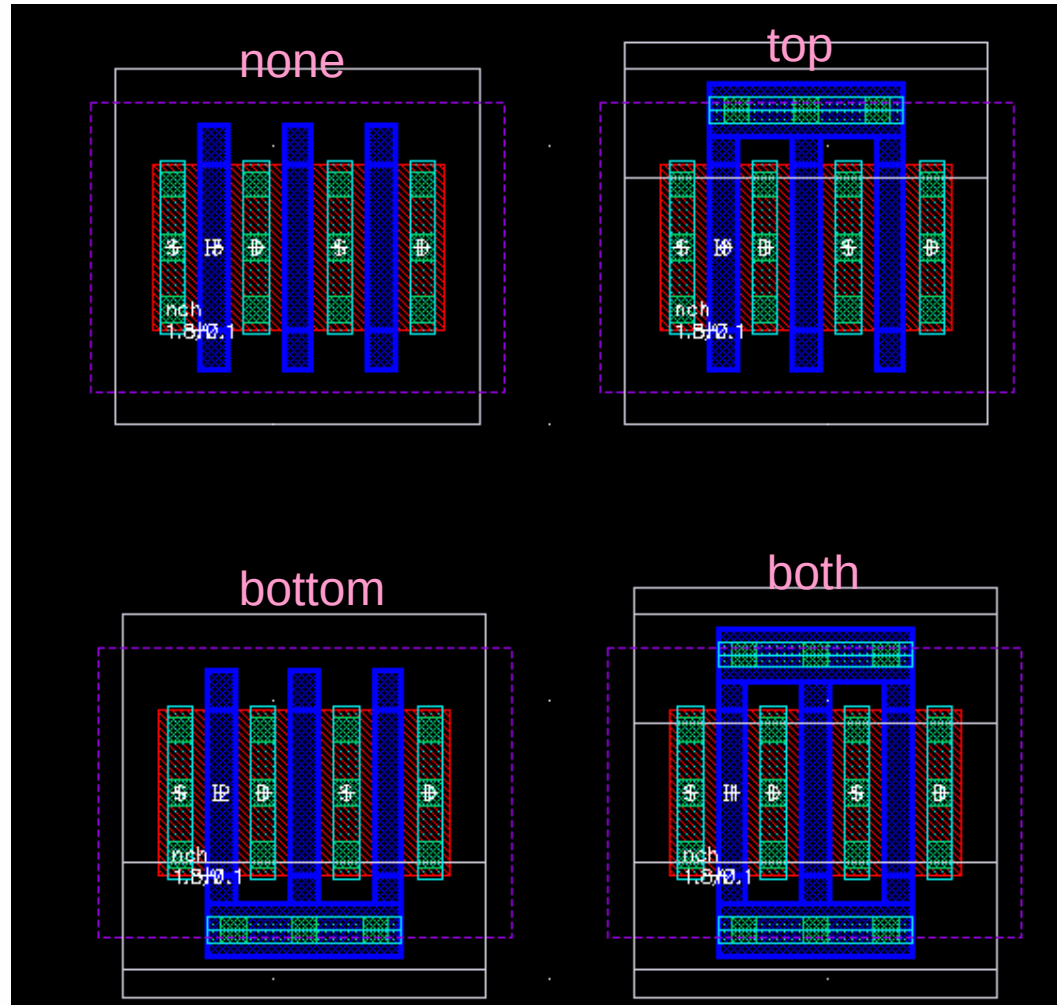


TSMC018 PDK Usage Guide

● The function of routePolydir

- Finger number=3. If customer use finger number=3 and turn on the routePolydir none/top/bottom/both direction. It will shows the following layout.

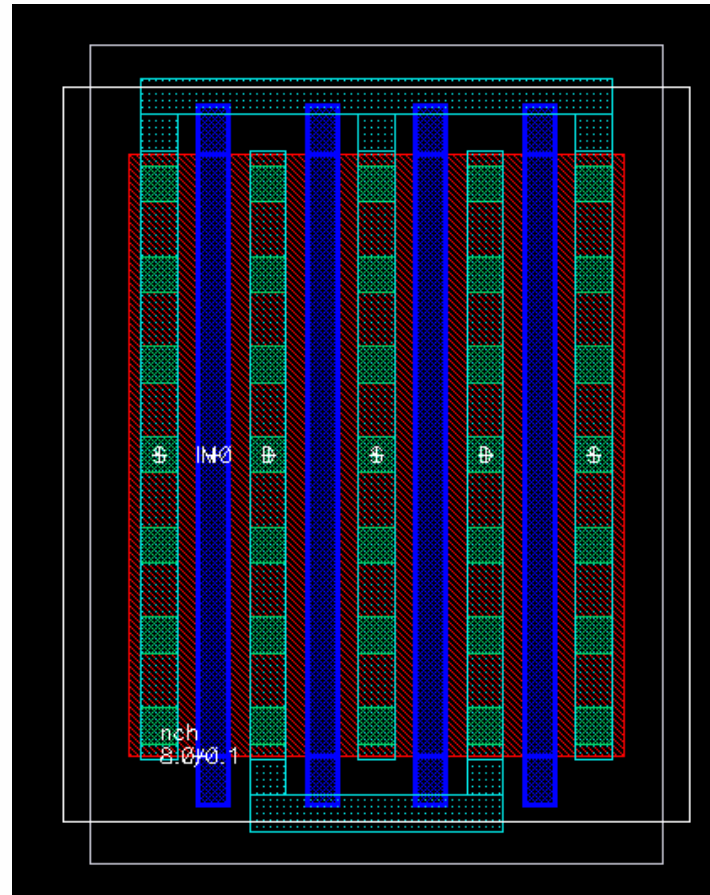
Finger=3



TSMC018 PDK Usage Guide

- The function of **route_Source_Drain**
 - The function is provided to drawing source and drain connection.

route_Source_Drain is *Both*

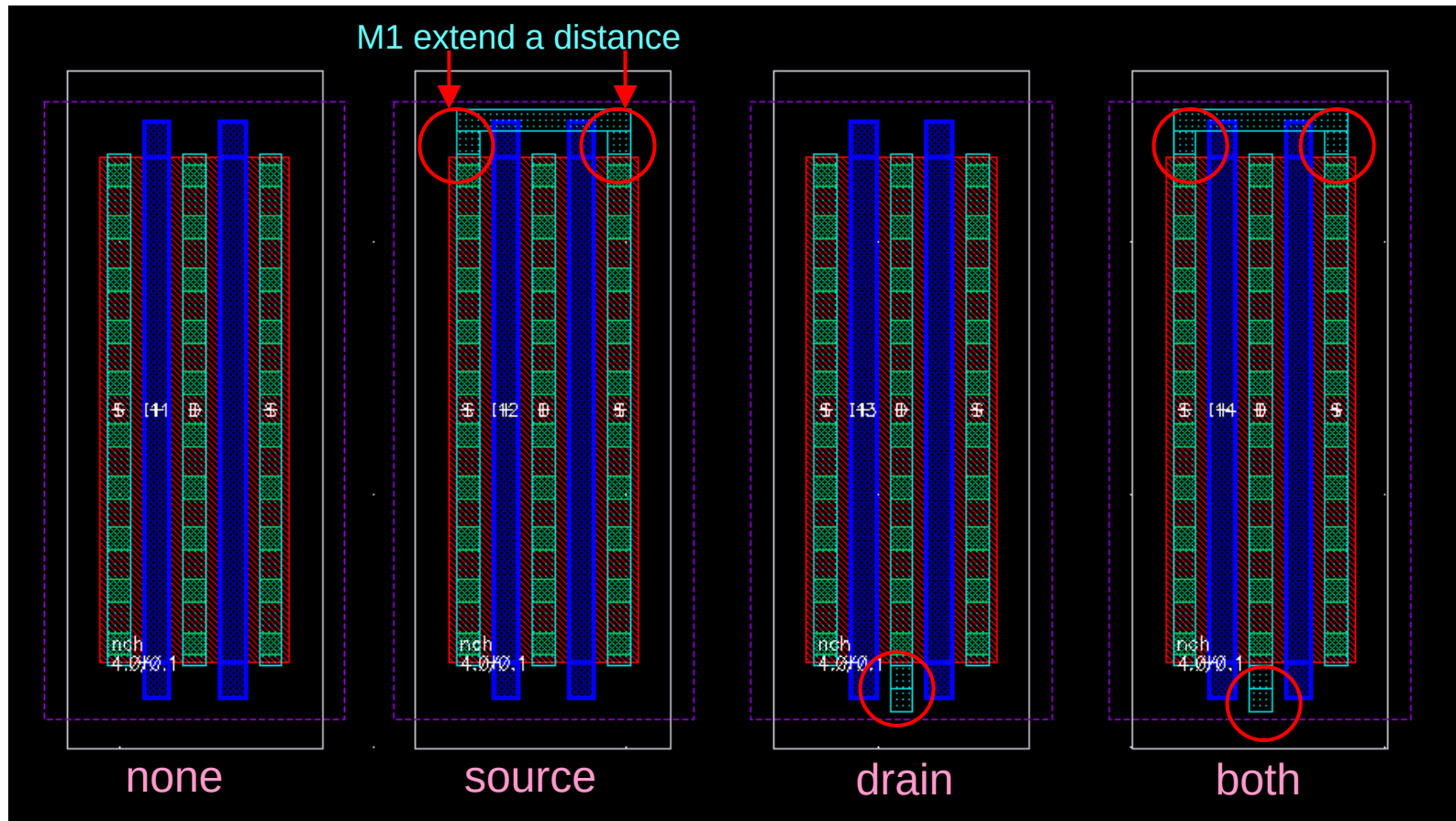


[Check here to back to MOS](#)

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● The function of `route_Source_Drain`

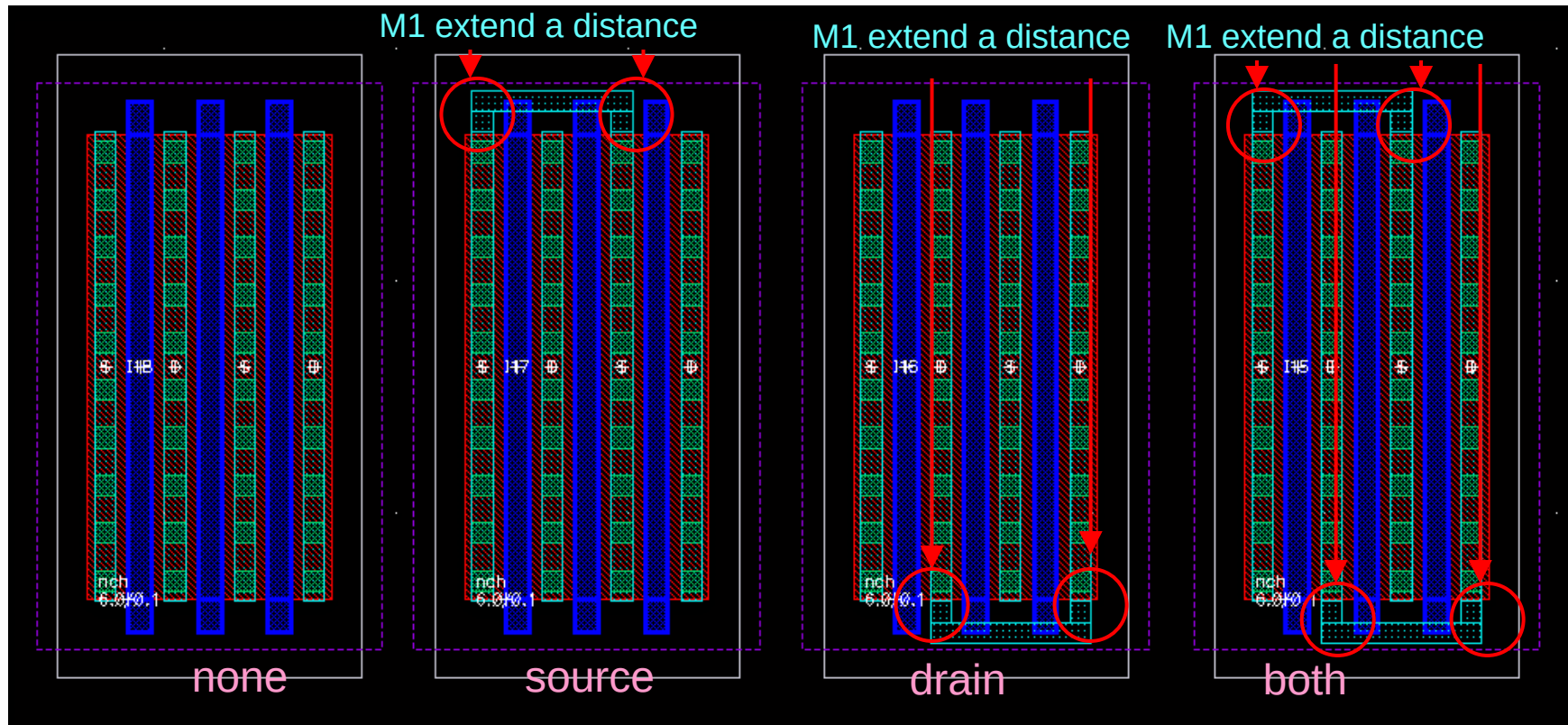
- The function is provided to drawing source and drain connection(it happens to $nf > 1$). User need to pay attention for the following case $Nf=2$; it means the mos has two source and one drain. If you turn on this function your source-drain metal layer will extend a distance.



TSMC018 PDK Usage Guide

● The function of `route_Source_Drain`

- The function is provided to drawing source and drain connection(it happens to $nf > 1$). User need to pay attention for the following case $Nf=3$; it means the mos has two source and one drain. If you turn on this function your source-drain metal layer will extend a distance.

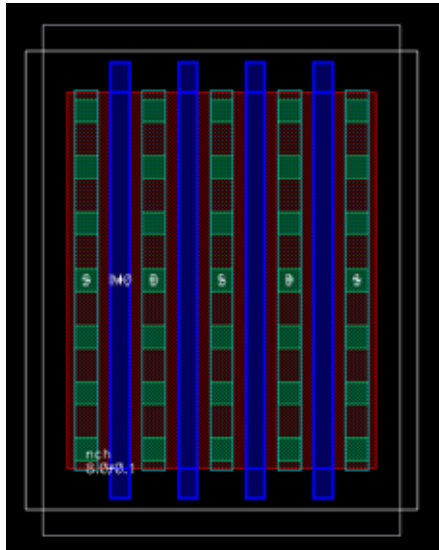


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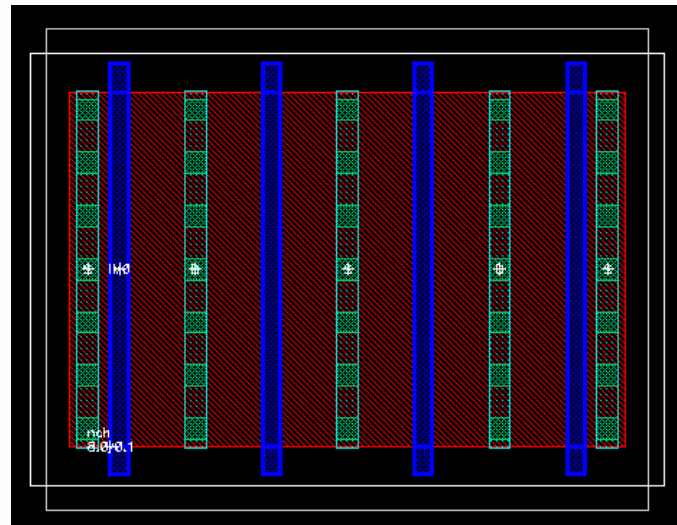
● The function of **fingers_SP_INC(M)**

- The function provide user to modify poly gate space. **Fingers_SP_INC(M)** is a increase value, it's not a distance between poly gate.

fingers_SP_INC(M) = 0



fingers_SP_INC(M) = 0.5



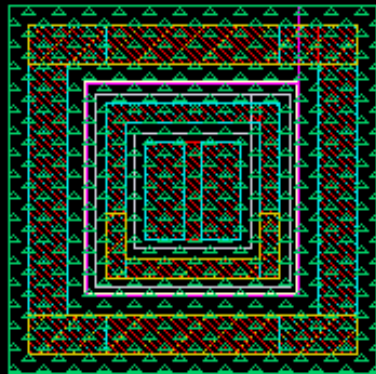
[Check here to back to MOS](#)

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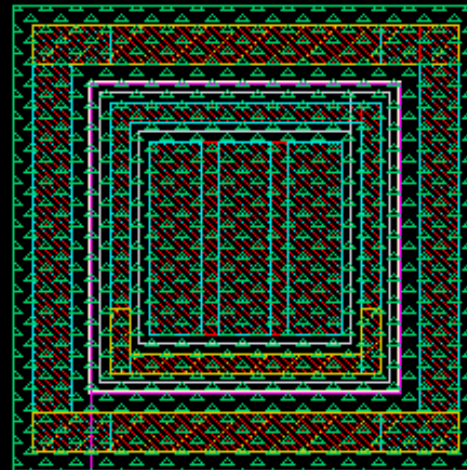
- The function of **BJT_type**

- There are three dimension of pnp and npn are provided in this PDK, user can use this function to choose those device layout.

BJT_type = 5X5



BJT_type = 10X10



[Check here to back to BJT](#)

TSMC018 PDK Usage Guide

- The function of **Total resistance(ohms)**, **Segment width(M)**, **Segment length(M)** and **Res_update_method**
 - In the resistance cell, we provide user two kinds of input method – **I_&_W** and **Rec_&_W** to modify the device resistance. When the user select **I_&_W** method, the input parameter will be **segment length(M)** and **segment width(M)**, the other one is **total resistance(ohms)** and **segment width(M)**.

select **I_&_W** method

CDF Parameter	Value	Display
Model name	rm1w	off
description	Metal 1 resistor	off
Total resistance(ohms)	447.000m Ohms	off
Segment width(M)	2u M	off
Segment length(M)	10u M	off
Multiplier	1	off
Rs(ohms/square)	0.0894	off
Hard_constrain	☐	off
Res_update_method	I_&_w	off

CDF Parameter	Value	Display
Model name	rm1w	off
description	Metal 1 resistor	off
Total resistance(ohms)	447.000m Ohms	off
Segment width(M)	2u M	off
Segment length(M)	10u M	off
Multiplier	1	off
Rs(ohms/square)	0.0894	off
Hard_constrain	☐	off
Res_update_method	Res_&_w	off

select **Rec_&_W** method

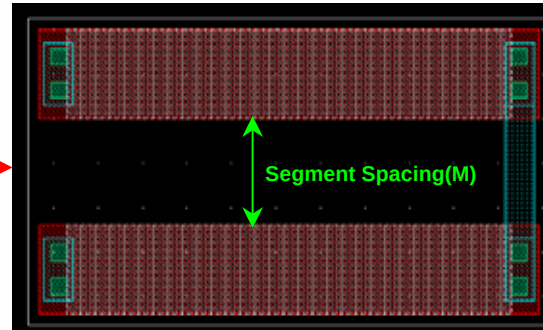
[Check here to back to Resistance\(2\)](#)

TSMC018 PDK Usage Guide

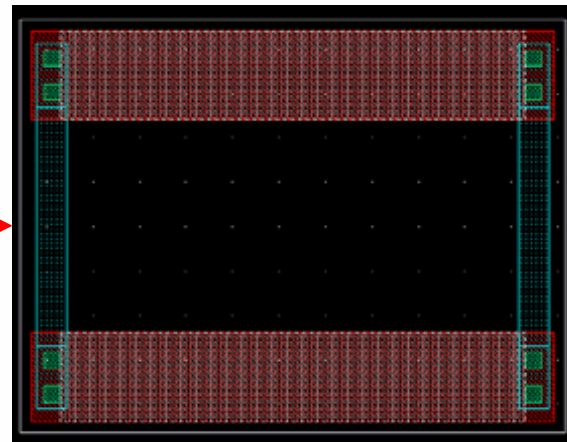
● The function of Resistor connection, Number of segment, Segment spacing(M)

- Number of segment provide user a function to Increment the number of segment resistance, user can use Resistor connection and Segment spacing(M) to modify connection type – **series** or **parallel** and segment spacing.

Number of segment = 2
Resistor connection = **series**
Segment Spacing(M) = 2.4u



Number of segment = 2
Resistor connection = **parallel**
Segment Spacing(M) = 4.8u

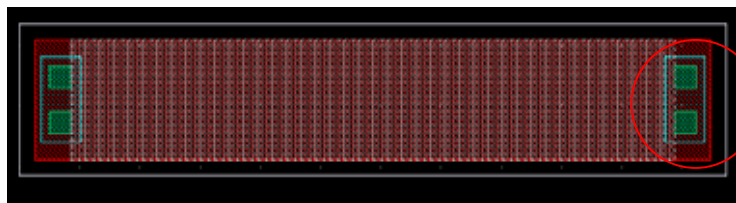


[Check here to back to Resistance\(2\)](#)

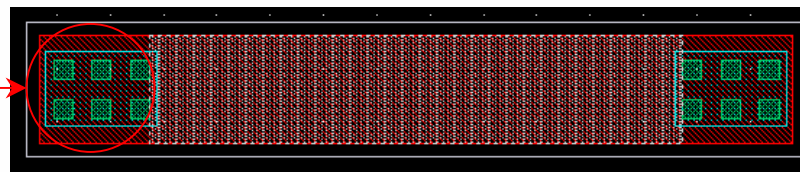
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- The function of **Cont columns**
 - This function provide user to modify the contact columns.

Cont columns= 1



Cont columns= 3



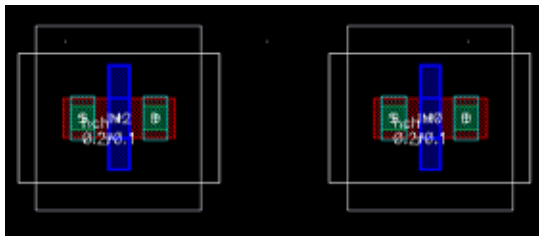
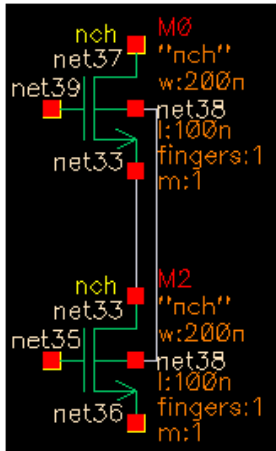
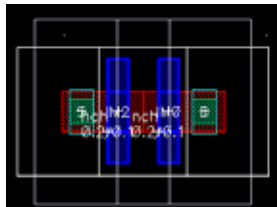
[Check here to back to Resistance\(2\)](#)

Appendix

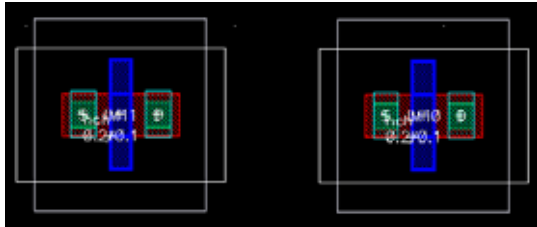
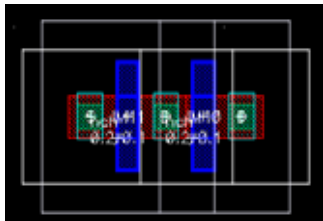
TSMC018 PDK Usage Guide

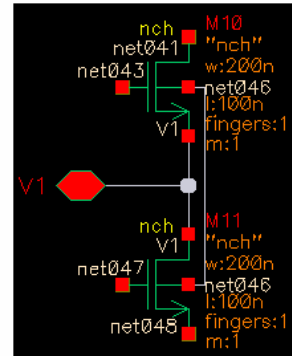
● Appendix A - Abutment

- To make the user understand this function, we describe more details and examples about abutment in this section. There are two point about this function is important, first this function only support MOS device in this PDK, second there are the same type MOS (ex: `nch` and `nch_18`, `pch` and `pch_25`) can be abutted only.
- When user abut two device, the terminal B must connects certainly in the layout view so it must be connected to the same net in the schematic view.
- The system will auto determine abutment type of the both device. User can't modify abutment type.
- The same devices abut case as show in below :(Case I)

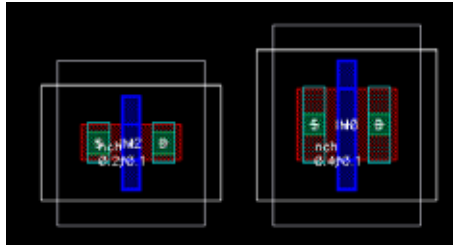
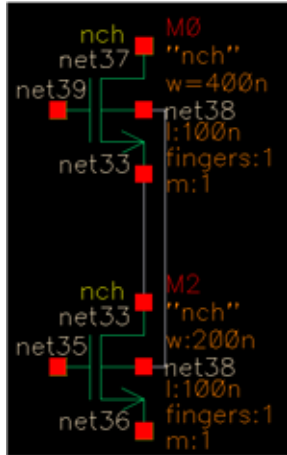
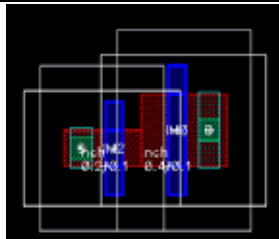
Device	nch (W=0.2u)	nch (W=0.2u)	The view in schematic
Abutment type	ShareLeftShape normal ShareRightShape same	ShareLeftShape same ShareRightShape normal	
Before Abutment			
After Abutment			

- Abutment ability lets you overlap two MOS, to create a connection between two sets of shapes overlapping each other. The two sets of shapes must include pins connected to the same net. (Case II)

Device	nch (W=0.2u)	nch (W=0.2u)	The view in schematic
Abutment type	ShareRightShape largeExt ShareLeftShape normal	ShareLeftShape sameExt ShareRightShape normal	
Before Abutment			
After Abutment			



- This case show out the different width device have been abutted. (Case III)

Device	nch (W=0.2u)	nch (W=0.4u)	The view in schematic
Abutment type	ShareLeftShape normal ShareRightShape small	ShareLeftShape large ShareRightShape normal	
Before Abutment			
After Abutment			

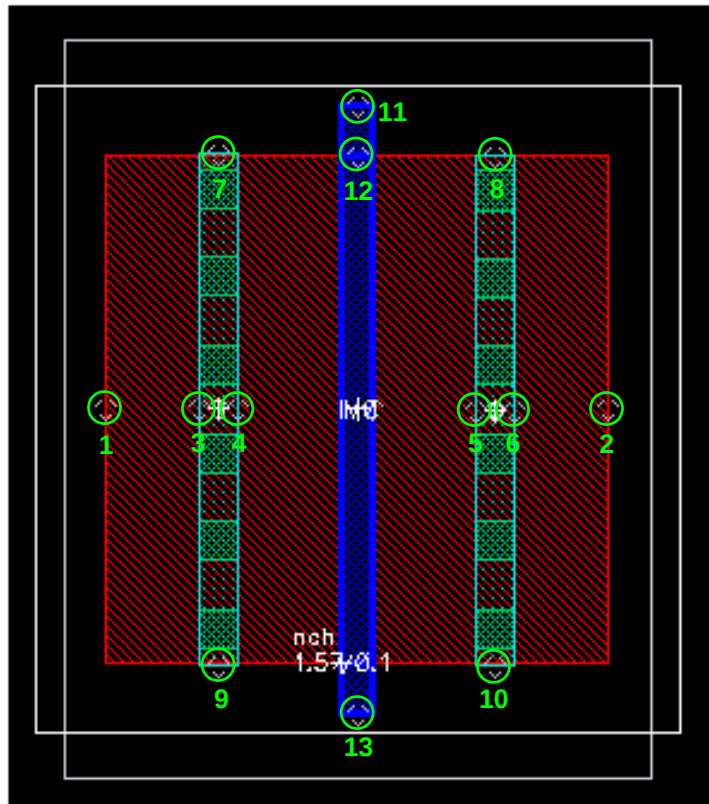
- This case show out that the same type MOS also can be abutted. (Case IV)

Device	nch (W=0.2u)	nch_25 (W=0.4u)	The view in schematic
Abutment type	ShareLeftShape normal ShareRightShape small	ShareLeftShape diff ShareRightShape normal	
Before Abutment			
After Abutment			

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● Appendix B – Stretch Handles

- This function lets user graphically change the value of those parameter for Pcell instances after user place them. The only one device MOS is a stretchable Pcell in this PDK.
- The system default is not show out the stretch handles, user must be enable the function manually. (Direct: in the layout view **Options** → **Display option** → **Stretch Handles**)



Stretch Handles number	Stretch direction
1	← →
2	← →
3, 5	←
4, 6	→
7, 8	↓
9, 10	↑
11	↑
13	↓
12	↑ ↓

- A example to show out the stretch case when the user stretch the handle 12.

